Name: _____

Question	1	2	3	4	5	*Total*
Points	6	6	6	6	6	30
Score						

1. **Some two-player game**

   Consider the two-player game with Player 1 as Row and Player 2 as Column players:

	L	R
U	$6, 2$	$2, 5$
M	$1, 0$	$5, 1$
D	$4, 3$	$4, 2$

   (a) (2 pts) Find all pure Nash equilibria.

   (b) (2 pts) Find all Pareto optimal outcomes.

   (c) (2 pts) Is the pure equilibrium Pareto optimal? If not, identify an outcome that dominates it and explain why this does not contradict unilateral stability.

Solution:

(a) Row's best responses are U to L and M to R. Column's best responses are R to U, R to M, and L to D. Their unique intersection is (M, R), with payoffs $(5, 1)$.

(b) The Pareto optimal outcomes are (U, L) with $(6, 2)$, (U, R) with $(2, 5)$, and (D, L) with $(4, 3)$. They trade off the two scores. The remaining outcomes are dominated: (M, L) and (M, R) by (U, L), and (D, R) by (D, L).

(c) No: (U, L) improves both payoffs over (M, R). But at (M, R), Row's unilateral change to U gives $2 < 5$, and Column's unilateral change to L gives $0 < 1$. Reaching the better outcome requires a joint change, which the Nash test does not consider.

2. **Three actions and a numerical candidate**

   Consider the following game:

	L	R
U	$5, 0$	$0, 4$
M	$1, 4$	$4, 0$
D	$3, 1$	$3, 3$

   (a) (3 pts) Find a Nash equilibrium in which Row uses M, D and Column uses both actions. Verify against the unused action U as well.

(b) (3 pts) Against Column's uniform strategy, compute Row's logit response when $e^{\lambda/2} = 2$. Is this response a best response in the Nash sense? Explain why the two concepts differ here.

Solution:

(a) Against uniform Column play, Row's pure payoffs are $\frac{5}{2}, \frac{5}{2}, 3$. Its current payoff is $\frac{5}{2}$, so its deviation incentive is $\delta_1 = \frac{1}{2}$, attained by D. Against the proposed Row strategy, Column obtains 2 from either action. Thus $\delta_2 = 0$ and $\epsilon_{\min} = \frac{1}{2}$.

(b) Let $q = P(L)$ and $s = P(M)$, with $P(D) = 1 - s$. The indifference conditions are

$$4 - 3q = 3, \qquad 4s + (1 - s) = 3(1 - s).$$

Hence $q = s = \frac{1}{3}$. The equilibrium is

$$p_1 = (0, \tfrac{1}{3}, \tfrac{2}{3}), \qquad p_2 = (\tfrac{1}{3}, \tfrac{2}{3}),$$

with payoffs $(3, 2)$. Row's unused U gives $\frac{5}{3} < 3$, while both used actions give 3. Column's actions both give 2, so every unilateral deviation is checked.

(c) Subtracting $\frac{5}{2}$ from the three expected utilities leaves $(0, 0, \frac{1}{2})$, so the logit weights are $(1, 1, 2)$. The response is $(\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$. It is not a Nash best response: its payoff is $\frac{11}{4} < 3$, obtainable by pure D. A finite-precision logit response assigns positive probability even to strictly inferior responses; a Nash best response puts weight only on payoff-maximizing actions.

3. **A matrix-game guarantee**

Consider the zero-sum game:

	L	C	R
U	3	2	1
M	0	1	5
D	2	3	1

(a) (1 pt) Find the maximin and minimax values in pures trategies. Is there a pure Nash equilibrium?

(b) (2 pts) Write the linear programs for both players' optimal mixed strategies.

(c) (3 pts) Row has an optimal strategy assigning positive probability to all three rows. Find it by equalizing the payoffs against all three columns. Verify that Column's uniform strategy is optimal and determine the value of the game. Explain why these checks certify optimality.

Solution:

(a) Row minima are $1, 0, 1$; column maxima are $3, 3, 5$. The pure maximin is 1 and the pure minimax is 3. There is no pure saddle point.

(b) For Row's probabilities x_U, x_M, x_D and Column's probabilities y_L, y_C, y_R, the programs are

$$\begin{aligned}
&\max\ v \\
&\text{s.t. } 3x_U + 2x_D \geq v, \quad 2x_U + x_M + 3x_D \geq v, \\
&\qquad x_U + 5x_M + x_D \geq v, \quad x_U + x_M + x_D = 1, \quad x \geq 0; \\
&\min\ w \\
&\text{s.t. } 3y_L + 2y_C + y_R \leq w, \quad y_C + 5y_R \leq w, \\
&\qquad 2y_L + 3y_C + y_R \leq w, \quad y_L + y_C + y_R = 1, \quad y \geq 0.
\end{aligned}$$

The payoff bounds v, w are unrestricted real variables.

(c) Equality of the first two column payoffs gives $x_U = x_M + x_D$, hence $x_U = \frac{1}{2}$. Equality of the first and third gives $2x_U - 5x_M + x_D = 0$. Using $x_M + x_D = \frac{1}{2}$ yields

$$x = (\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{4}).$$

It gives Row exactly 2 against each column. Against uniform Column play, each row gives 2, since every row sum is 6. Thus Row guarantees 2 and Column holds Row to at most 2; both strategies are optimal and the value is 2. Expected payoffs against mixed deviations are averages of the checked pure-action payoffs.

4. **Adjusting a recommendation device**

Row and Column have the following game:

	L	C	R
U	$4,3$	$2,4$	$0,0$
M	$2,0$	$4,4$	$3,3$
D	$3,5$	$0,0$	$5,4$

A device privately recommends actions using $p(U,L) = t, p(U,C) = \frac{1}{3} - t, p(M,C) = p(M,R) = p(D,L) = p(D,R) = \frac{1}{6}$, with all other probabilities zero and $0 \leq t \leq \frac{1}{3}$. You may use the fact that Row's recommendations M, D and Column's recommendations C, R are obedient throughout this interval.

(a) (3 pts) Derive both obedience inequalities for Row when recommended U and both for Column when recommended L. Which t give a correlated equilibrium?

(b) (3 pts) Within this family, find the correlated equilibrium maximizing Row's expected payoff, and give both players' expected payoffs.

Solution:

(a) Using joint probabilities, the four inequalities are

$$\begin{array}{rcll} \frac{2}{3}+2t & \geq & \frac{4}{3}-2t & (U \to M), \\ \frac{2}{3}+2t & \geq & 3t & (U \to D), \\ 3t+\frac{5}{6} & \geq & 4t & (L \to C), \\ 3t+\frac{5}{6} & \geq & \frac{2}{3} & (L \to R). \end{array}$$

They give $t \geq \frac{1}{6}$, $t \leq \frac{2}{3}$, $t \leq \frac{5}{6}$, and $t \geq -\frac{1}{18}$. With the prescribed interval, the CE range is $\frac{1}{6} \leq t \leq \frac{1}{3}$.

(b) Row receives U with probability $\frac{1}{3}$, so conditional probabilities of L, C, R are $(\frac{1}{4}, \frac{3}{4}, 0)$. Obeying gives $\frac{5}{2}$; switching to M gives $\frac{7}{2}$, while switching to D gives $\frac{3}{4}$. The best deviation is therefore M, with conditional gain 1.

(c) Expected payoffs under obedience are

$$U_1 = 4t + 2(\tfrac{1}{3} - t) + \tfrac{4+3+3+5}{6} = \tfrac{19}{6} + 2t, \qquad U_2 = 4 - t.$$

Row's payoff increases with t, so the optimum is $t = \frac{1}{3}$. The resulting payoffs are $(\frac{23}{6}, \frac{11}{3})$.

5. **Three possible commitments**

Row commits to probabilities (a, b, c) on U, M, D. Column observes these probabilities, but not the sampled row, and best responds:

	L	R
U	$2, 3$	$4, 0$
M	$5, 0$	$1, 3$
D	$4, 1$	$3, 2$

(a) (4 pts) Write and solve the two leader LPs, one per follower action. Find the strong Stackelberg equilibrium and both payoffs. Explain meaning of all parts of the linear program. To solve the programs, it suffices to check feasible pure commitments and the two mixtures $(\frac{1}{2}, \frac{1}{2}, 0)$ and $(\frac{1}{4}, 0, \frac{3}{4})$.

(b) (2 pts) Does a weak Stackelberg equilibrium exist? Justify your answer and give the supremum of the leader's payoff under weak tie-breaking.

Solution:

(a) Column receives $3a + c$ from L and $3b + 2c$ from R. Thus L is best when $3a \geq 3b + c$, R when $3a \leq 3b + c$; equality gives a tie.

(b) Both programs include $a + b + c = 1$ and $a, b, c \geq 0$:

$$\begin{aligned} LP_L: &\quad \max(2a + 5b + 4c) \quad \text{s.t. } 3a \geq 3b + c, \\ LP_R: &\quad \max(4a + b + 3c) \quad \text{s.t. } 3a \leq 3b + c. \end{aligned}$$

(c) For LP_L, feasible pure U gives 2, and each displayed mixture gives $\frac{7}{2}$. For LP_R, pure M, D give $1, 3$, and the mixtures give $\frac{5}{2}, \frac{13}{4}$. The strong value is therefore $\frac{7}{2}$. One SSE is $(a, b, c) = (\frac{1}{2}, \frac{1}{2}, 0)$ with response L, giving payoffs $(\frac{7}{2}, \frac{3}{2})$. Every mixture of the two displayed commitments is also optimal with response L; at each, L gives the leader strictly more than R.

(d) No. Write $d = 3a - 3b - c$. Whenever L is uniquely best, $d > 0$ and the leader's payoff is $\frac{7}{2} - \frac{1}{2}d < \frac{7}{2}$. At a tie the weak rule chooses R, whose branch never exceeds $\frac{13}{4}$. But $(a, b, c) = (\frac{1}{2} + \eta, \frac{1}{2} - \eta, 0)$, $0 < \eta < \frac{1}{2}$, induces L uniquely and gives $\frac{7}{2} - 3\eta$. Hence the supremum is $\frac{7}{2}$, approached but never attained.